

Riemannian TTN reproduction

Fashion-MNIST · Independent implementation · 3 October 2026

This experiment tests the Adam–RADAM comparison in Willner et al., arXiv:2609.00870, Figures 5–6. The compression advantage is reproduced qualitatively. The reported equality of final classification accuracy is not reproduced under the reconstruction choices below. The experiment establishes a working manifold-optimization baseline; it supplies no evidence yet about logical reasoning.

Classification after 30 epochs

Optimizer	Test accuracy	Test loss	Train time
Adam	89.9%	0.1861	191 s
RADAM, printed	84.59%	0.2797	342 s
RADAM, corrected	86.8%	0.2449	339 s

The paper reports approximately 89% accuracy for all optimizers. These are single-seed results, with identical initial weights and minibatch ordering. Training time excludes initialization and evaluation. Apple MPS timings are not directly comparable with the paper’s NVIDIA A30 timings.

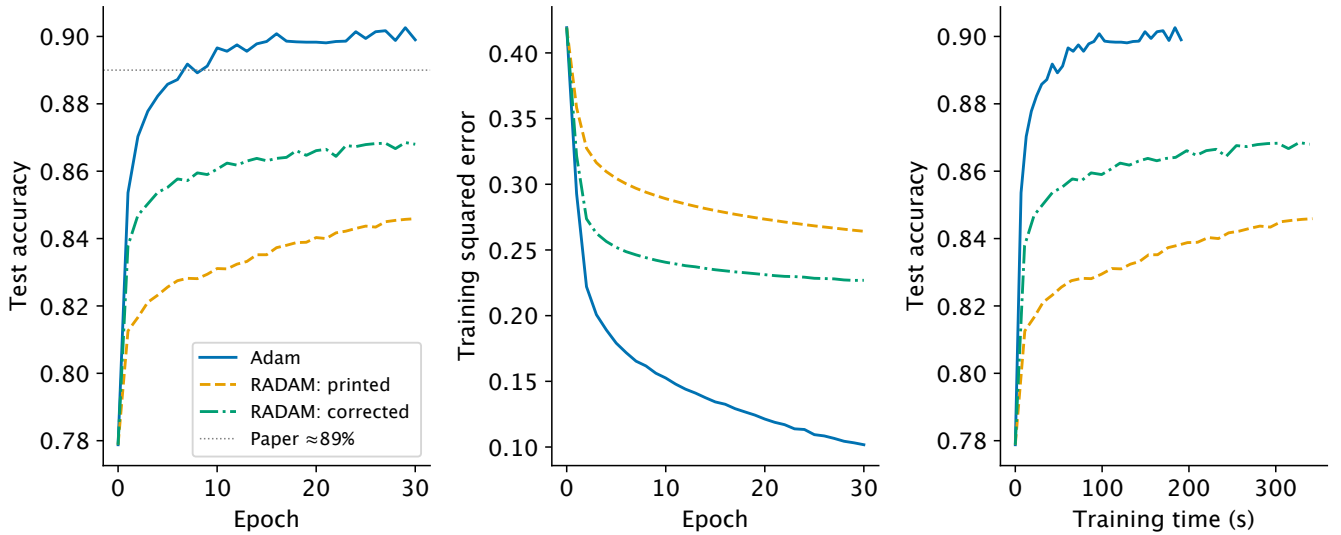


Figure 1: Test accuracy, online minibatch training loss, and test accuracy against training time. The dotted reference is the paper’s approximate final accuracy. Epoch zero uses a full training-set loss, before optimization.

The experiment uses the complete official Fashion-MNIST split, 16×16 images, a balanced TTN with maximum bond 16, and the paper’s training budget and learning rates. Both second-moment formulas are retained because the printed pseudocode and conventional Adam differ. The comparison is a single independent run with explicit reconstruction choices, rather than a verification using author code.

The qualitative compression result is useful for selecting a baseline for the proposed tensor logic layer. It does not establish better reasoning, and the classification discrepancy remains unresolved.

Compression and the precision control

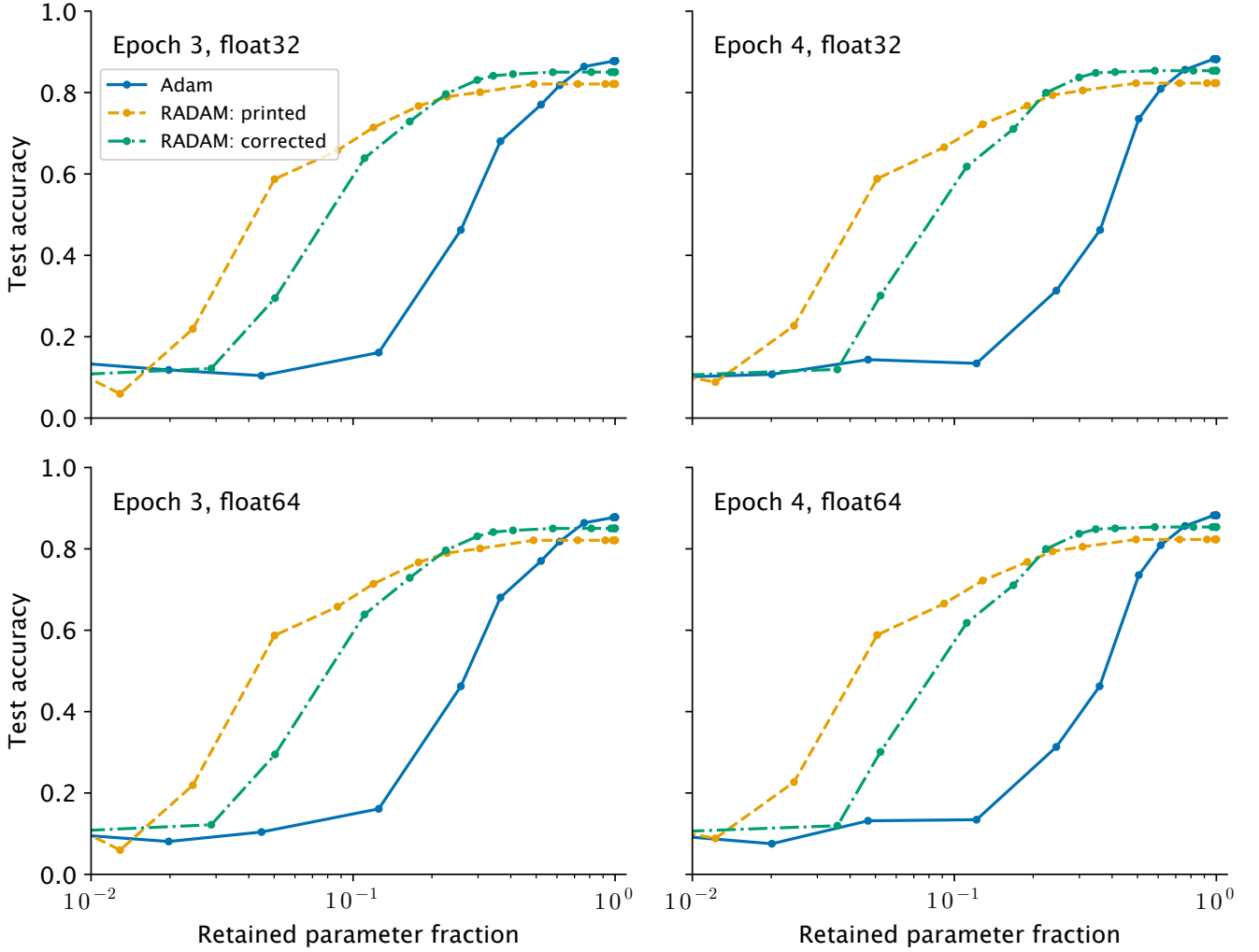


Figure 2: Accuracy after Schmidt compression at epochs 3 and 4. Each point is a measured retained parameter count; lines connect sampled tolerances. Float32 and float64 use the same CPU backend. The grid includes zero-tolerance controls.

The compression sweep follows the discarded squared-Schmidt-weight criterion in Scharf et al., arXiv:2604.25755, Section 2.4.3. After QR canonicalization, a depth-first SVD sweep visits the internal edges. Physical and class legs are fixed. The tolerance is scaled by the original full-tensor norm squared. Norm calculation uses explicit scaling to avoid avoidable overflow before the norm itself leaves the numeric range. The paper does not give its exact sweep or tolerance grid.

At epoch 3, the full tensor norm is approximately 1.94×10^{18} for Adam and 3.68×10^2 for corrected RADAM. At epoch 4, Adam’s norm is about 6.99×10^{20} . Zero-tolerance float64 canonicalization preserves Adam’s predictions. Its loss of accuracy under truncation nevertheless persists in float64. In this run, numerical overflow alone cannot explain the compression gap. The growing norm is also a change of the represented tensor: exact internal gauge changes preserve that norm.

Implementation and unresolved details

The balanced binary TTN has 256 input sites, local feature map $\varphi(x) = \frac{(1,x)^T}{\sqrt{1+x^2}}$, maximum bond dimension 16, and a free 10-class root. Its 274,944 parameters are trained on all 60,000 official training images and evaluated on all 10,000 test images. Loss is the batch mean of the sum of squared one-hot class errors. Batch size is 256; learning rates are 0.001 for Adam and 0.03 for RADAM.

Nonroot cores satisfy $B^T B = I$. RADAM uses horizontal projection $G \rightarrow G - B(B^T G)$, one scalar second moment per core, momentum transport by projection, and the POGO correction $M \rightarrow M(1.5I - 0.5M^T M)$ from Javaloy and Vergari, arXiv:2602.14656. Float32 training follows the paper; initialization uses float64.

Algorithm 1 prints $(1 - \beta_1)$ in the second-moment update, where conventional Adam uses $(1 - \beta_2)$. Both interpretations are evaluated, without bias correction. “Corrected” denotes our sensitivity variant, not an author-confirmed erratum. We did not tune learning rates to match the test accuracy.

The initialization follows the reduced-covariance construction in Stoudenmire, arXiv:1801.00315, including the norms of every branch traced out. We choose bilinear antialiased resizing, row-major pixels in $[0, 1]$, maximum allowed ranks, and a ridge-fitted root. The relative ridge coefficient is 10^{-6} . The target paper does not fully specify these choices. They limit the strength of a numerical comparison.

Square-core numerical limitation

The reported run evaluates the horizontal projection literally in float32. The bottom two levels have square cores, whose exact quotient space is a point. Projection roundoff can nevertheless be amplified by the adaptive denominator. Thus these measurements use approximate finite-precision geometry. They should not be treated as an exact-manifold control.

The implementation now offers `--square-projection freeze`, which leaves these cores unchanged and is covered by a regression test. The full training results have not been repeated with that option. This distinction is explicit in the CLI and run documentation; the original results and source snapshot are retained.

Checks and implications for the proposed logic layer

Eleven targeted tests cover dense contraction and differentiation, tangent projections, the POGO error order, the printed RADAM update, reduced covariances, canonicalization, truncation error bounds, packed inference, large representable float32 norms, frozen square cores, resume rejection, and failure logging. A separate end-to-end check found bitwise identical parameters after continuous and resumed training for both Adam and printed RADAM. All source papers were downloaded and inspected; PDF hashes are recorded.

The next study should preserve this compression control when replacing the classification head with a tensor logic layer. Compare at matched predictive accuracy or matched reasoning success before attributing a compression advantage to improved reasoning. First resolve the root initialization and optimizer conventions, then test length generalization and rule composition on the chosen real-world benchmark. This reproduction does not support a reasoning claim by itself.

Reproduction entry point: experiments/ttn/README.md in pdft-benchmarks. Code, environment pins, downloaded-source hashes, raw JSON measurements, checkpoints, and vector figures accompany this report. The run uses seed 0; no uncertainty across seeds has been estimated.